

Part 3: Learn Lit.js

Goal: understand the LUX Lab widget component line by line, and see it render on the LUX page.

Where this sits: Part 1 Setup → Part 2 Build Experience → **Part 3 Learn Lit.js** → Part 4 Connect to Server

Steps

1. **Recap from Part 2**
   - LUX page live, `<aiux-lux-lab>` placed on it
2. **What is Lit, in one minute**
   - Web components: your own HTML tags
   - Lit: small library to build them
   - AngularJS widget → Lit widget
3. **Open `widgets/aiux-lux-lab/index.js`**
4. **Imports**
   - `html`, `css` from `lit`
   - `customElement` from `lit/decorators.js`
   - `AIUXWidgetElement` and `decorators` from `aiux-components-core`
   - `i18n` from `aiux-services`
5. **Widget decorators, the widget's ID card**
   - `@customElement('aiux-lux-lab')`: the tag name
   - `@name`, `@description`: how it shows in the Builder
   - `@category('custom')`: grouping
   - `@server('./server-script.js')`: link to the server script (Part 4)

- `@bestFor`, `@chatCompatible`, `@discoverable`: metadata for AI and discovery (not used in this series)

6. **The class**
   - `extends AIUXWidgetElement`
   - Adds `this.data` and server connection on top of Lit
   - Page vs widget: `AIUXElement` vs `AIUXWidgetElement`
7. **`static properties`**
   - `_title`, `_greeting` with `type: String`
   - `state: true` = internal value
   - Change a property → widget re-renders
8. **`static styles`**
   - `:host`: styles the widget tag itself
   - `.ainow`: our own class, green heading
9. **Shadow DOM, quick demo**
   - Styles stay inside the widget
   - Toggle `createRenderRoot() { return this; }` and compare
10. **`constructor()`**
    - `super()` first
    - Default values for title and greeting
11. **`render()`**
    - Read values: `this.data?.title || this._title`
    - `?.` = safe when data is not there yet
    - `html` template with `${}` placeholders
12. **Conditional rendering**
    - Incident found → number + description
    - Not found → error or "No incidents found"
13. **`i18n.getMessage()` for labels**
14. **Build, deploy, open `/lux-lab`**
15. **Checkpoint**
    - Widget shows default title and greeting
    - Timestamp and incident empty: no server yet

Next: Part 4, Connect to Server. We fill `this.data` from the instance.

AINOW
Subscribe for more ServiceNow + AI deep dives
youtube.com/@aibytusharm
Scan to subscribe

